

Section Id :	64065364143
Section Number :	15
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	12
Number of Questions to be attempted :	12
Section Marks :	45
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653134175
Question Shuffling Allowed :	No

Question Number : 420 Question Id : 640653904485 Question Type : MCQ Calculator : Yes
Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533045499. ✓ YES

6406533045500. ✗ NO

Sub-Section Number :	2
Sub-Section Id :	640653134176
Question Shuffling Allowed :	No

Question Id : 640653904497 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (421 to 423)

Question Label : Comprehension

Calculate the Shapley value of the following game:

$$N = \{1, 2, 3\} \quad v(1) = 6 \quad v(2) = 12 \quad v(3) = 18 \\ v(1, 2) = 12 \quad v(1, 3) = 12 \quad v(2, 3) = 24 \\ v(1, 2, 3) = 48 \quad v(\emptyset) = 0$$

Suppose the Shapley value is (s_1, s_2, s_3) ,
then answer the given subquestions

Sub questions

Question Number : 421 Question Id : 640653904498 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

$$s_1 = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

11

Question Number : 422 Question Id : 640653904499 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

$$s_2 = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

14

Question Number : 423 Question Id : 640653904500 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

$$s_3 = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

23

Question Id : 640653904501 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (424 to 426)

Question Label : Comprehension

A joint enterprise of three partners, 1, 2, and 3, consists of a travel agency, a car rental agency, a tour company, a souvenir shop, and a delivery service. The owner of the car rental agency, 1, forms a partnership with the owner of the travel agency, 2. Later the two purchase a souvenir shop, which they let their wives run. The owner of the tour company, 3, enters into a partnership with 2 and the two open a delivery service. Now 3 enters into the comprehensive partnership and the enterprise is the operation of all these businesses together.

We need to remember that $v(S)$ is the price that coalition S can achieve in the market for all the assets that belong either to it or to its sub-coalitions. We shall describe the game in terms of the coalition function:

$$\begin{aligned} v(1) &= 20 & v(2) &= 30 & v(3) &= 10 & v(1, 2) &= 40 \\ v(1, 3) &= 0 & v(2, 3) &= 30 & v(1, 2, 3) &= 60 & v(\phi) &= 0 \end{aligned}$$

At a certain stage the joint enterprise starts losing money and the partners decide to liquidate it.

Calculate how much each of the partners will get from the sale of the whole enterprise, if they carry out the division according to the procedure for dissolving a partnership.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 424 Question Id : 640653904502 Question Type : SA Calculator : None Correct Marks : 2

Question Label : Short Answer Question

Partner 1 will get _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

16.6 to 16.8

Question Number : 425 Question Id : 640653904503 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

Partner 2 will get _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

36.6 to 36.8

Question Number : 426 Question Id : 640653904504 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

Partner 3 will get _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

6.6 to 6.8

Sub-Section Number : 3

Sub-Section Id : 640653134177

Question Shuffling Allowed : No

Question Id : 640653904486 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (427 to 428)

Question Label : Comprehension

Consider the following game:

$$N = \{1, 2, 3, 4\} \quad v(1) = v(2) = 0 \quad v(3) = v(4) = 1$$

$$v(\phi) = 0$$

$$v(1, 2) = 0 \quad v(2, 3) = 1$$

$$v(1, 3) = 1 \quad v(2, 4) = 1$$

$$v(1, 4) = 1 \quad v(3, 4) = 2$$

$$v(1, 2, 3) = 1 \quad v(2, 3, 4) = 2$$

$$v(1, 2, 4) = 1 \quad v(1, 3, 4) = 2 \quad v(1, 2, 3, 4) = 2$$

Based on the above data, answer the given subquestions

Sub questions

Question Number : 427 Question Id : 640653904487 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the null players in this game

Options :

6406533045501. ✓ 1 and 2

6406533045502. ✗ 1 and 3

6406533045503. ✗ 2 and 3

6406533045504. ✗ 2 and 4

6406533045505. ✗ 1 and 4

6406533045506. ✗ 3 and 4

Question Number : 428 Question Id : 640653904488 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

Choose the correct alternative.

Options :

6406533045507. ✗ There is a single pair of symmetric players in this game

6406533045508. ✓ If value of $v(2)$ changes to $v(2)=2$, players (1, 2) are no more null players

6406533045509. ✗ There is a single pair of symmetric players in this game & If value of $v(2)$ changes to $v(2)=2$, players (1, 2) are no more null players

6406533045510. ✗ None

Question Id : 640653904509 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Calculator : None

Question Numbers : (429 to 430)

Question Label : Comprehension

A machine can be in two possible states viz. Good (G) and Bad (B). It is in the Good state 90% of the

time. An item produced by the machine is Defective (D), 1% of the time if it is Good and 10% of the time if it is Bad.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 429 Question Id : 640653904510 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

What is the probability of the item being defective

Options :

6406533045527. ✖ 0.009

6406533045528. ✖ 0.09

6406533045529. ✔ 0.019

6406533045530. ✖ 0.19

Question Number : 430 Question Id : 640653904511 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

What is the probability that the machine is good if the item is defective

Options :

6406533045531. ✖ 0.047

6406533045532. ✔ 0.47

6406533045533. ✖ 0.0047

6406533045534. ✖ 0.19

Question Id : 640653904517 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (431 to 432)

Question Label : Comprehension

Five bidders with private valuations uniformly distributed between 0 and 1 participate in a sealed bid first price auction. Assuming that everyone is playing the symmetric equilibrium bidding strategy answer the given subquestions:

Sub questions

Question Number : 431 Question Id : 640653904518 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

Optimal bid for a bidder with valuation 0.6 will be

Options :

6406533045551. ✖ 0.4

6406533045552. ✓ 0.48

6406533045553. ✗ 0.5

6406533045554. ✗ 0.55

Question Number : 432 Question Id : 640653904519 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

The expected revenue of the seller in this auction scenario will be

Options :

6406533045555. ✗ 4/5

6406533045556. ✗ 3/5

6406533045557. ✓ 2/3

6406533045558. ✗ 2/5

Question Id : 640653904520 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (433 to 434)

Question Label : Comprehension

Consider the following game

	L	R
U	1,1	0,0
D	0,0	0,0

Based on the above data, answer the given subquestions

Sub questions

Question Number : 433 Question Id : 640653904521 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

Consider following two statements about this game and choose correct alternative

P: D is weekly dominated by U

Q: L is weekly dominated by R

Options :

6406533045559. ✓ Only P is correct

6406533045560. ✗ Only Q is correct

6406533045561. ✗ Both P and Q are correct

6406533045562. ✖ None is correct

Question Number : 434 Question Id : 640653904522 Question Type : MCQ Calculator : Yes Correct Marks : 1

Question Label : Multiple Choice Question

Consider following two statements about this game and choose the correct alternative

P: (D, R) is a Nash equilibrium.

Q: Elimination of weakly dominated strategies also eliminates (D, R), a Nash equilibrium.

Options :

6406533045563. ✖ Only P is correct

6406533045564. ✖ Only Q is correct

6406533045565. ✔ Both P and Q are correct

6406533045566. ✖ None is correct

Sub-Section Number :

4

Sub-Section Id :

640653134178

Question Shuffling Allowed :

No

Question Id : 640653904512 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (435 to 438)

Question Label : Comprehension

Given the following preference structure, find a stable matching system using the Gale–Shapley algorithm, when the women propose:

Women: A, B, C.				Men: a, b, c, d.			
	A	B	C		A	B	C
a	1	1	0	a	0	0	1
b	2	0	1	b	3	1	2
c	3	2	3	c	2	1	0
d	4	0	2	d	2	3	1

Based on the above data, answer the given subquestions

Sub questions

Question Number : 435 Question Id : 640653904513 Question Type : MCQ Calculator : Yes Correct Marks : 1

Question Label : Multiple Choice Question

Mr. a will be paired with

Options :

6406533045535. ✖ A
6406533045536. ✖ B
6406533045537. ✖ C
6406533045538. ✔ None

Question Number : 436 Question Id : 640653904514 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

Mr. b will be paired with

Options :

6406533045539. ✖ A
6406533045540. ✖ B
6406533045541. ✔ C
6406533045542. ✖ None

Question Number : 437 Question Id : 640653904515 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

Mr. c will be paired with

Options :

6406533045543. ✖ A
6406533045544. ✔ B
6406533045545. ✖ C
6406533045546. ✖ None

Question Number : 438 Question Id : 640653904516 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

Mr. d will be paired with

Options :

6406533045547. ✔ A
6406533045548. ✖ B
6406533045549. ✖ C
6406533045550. ✖ None

**Question Id : 640653904530 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Calculator : None**

Question Numbers : (439 to 441)

Question Label : Comprehension

There are 3 firms that are considering entering a new market. The payoff for each firm that enters is $\frac{150}{n}$ where n is the number of firms that enter. The cost of entering is 62.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 439 Question Id : 640653904531 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

How many pure strategy equilibria are there in this game?

Options :

6406533045591. ✖ 1

6406533045592. ✖ 2

6406533045593. ✔ 3

6406533045594. ✖ 4

Question Number : 440 Question Id : 640653904532 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

In one of these pure strategy NEs, all three players enter the market

Options :

6406533045595. ✖ TRUE

6406533045596. ✔ FALSE

Question Number : 441 Question Id : 640653904533 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

There is a symmetric mixed strategy equilibrium where all three players enter with the same probability (say p). Find the value of p

Options :

6406533045597. ✖ 1/5

6406533045598. ✖ 3/5

6406533045599. ✔ 4/5

6406533045600. ✖ 2/5

Sub-Section Number :

5

Sub-Section Id :

640653134179

Question Shuffling Allowed :

No

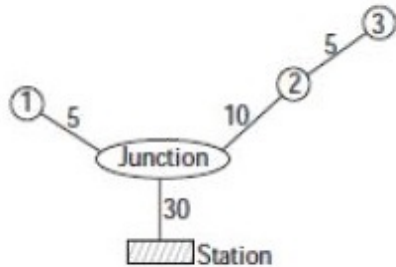
Question Id : 640653904505 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (442 to 444)

Question Label : Comprehension

A cable TV station wants to connect three new customers to its network.

The connections to the network are described in the following tree graph:



Calculate the distribution of the costs among the three players according to the Shapley value.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 442 Question Id : 640653904506 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 1 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

15

Question Number : 443 Question Id : 640653904507 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 2 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

15

Question Number : 444 Question Id : 640653904508 Question Type : SA Calculator : None
Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 3 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

20

Sub-Section Number :	6
Sub-Section Id :	640653134180
Question Shuffling Allowed :	No

Question Id : 640653904489 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

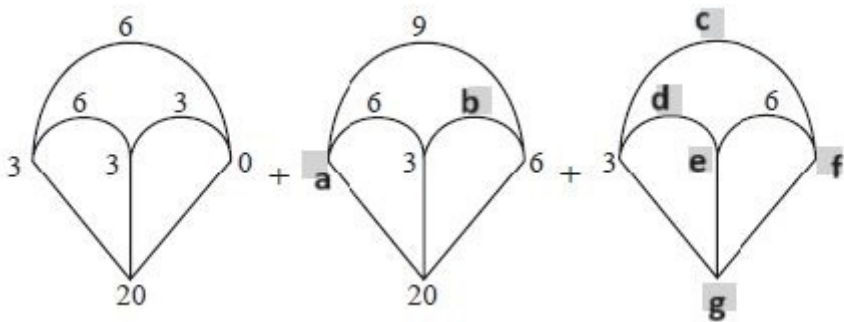
Question Numbers : (445 to 451)

Question Label : Comprehension

Consider the game $(N; v)$:

$$\begin{aligned} N &= \{1, 2, 3\} & v(1) &= 9 & v(2) &= 6 & v(3) &= 12 & v(\phi) &= 0 \\ & & v(1, 2) &= 15 & v(1, 3) &= 24 & v(2, 3) &= 18 \\ & & v(1, 2, 3) &= 60 \end{aligned}$$

The game is split into the three games below, whose sum is the original game. Supply the missing values.



Based on the above data, answer the given subquestions

Sub questions

Question Number : 445 Question Id : 640653904490 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

a= _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 446 Question Id : 640653904491 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

b= _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

9

Question Number : 447 Question Id : 640653904492 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

c= _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

9

Question Number : 448 Question Id : 640653904493 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

d= _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 449 Question Id : 640653904494 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

e=_____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 450 Question Id : 640653904495 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

f=_____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Question Number : 451 Question Id : 640653904496 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

g=_____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

20

Question Id : 640653904523 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (452 to 457)

Question Label : Comprehension

Each one of two political parties can choose to buy time on commercial radio shows to broadcast negative ad campaigns against their rival. These choices are made simultaneously. Due to government regulation, it is forbidden to buy more than 2 hours of negative campaign time so that each party cannot choose an amount of negative campaigning above 2 hours. Given a pair of choices (a_1, a_2) , the payoff of party i is given by the following function: $v_i(a_1, a_2) = a_i - 2a_j + a_i a_j - (a_i)^2$. Write the normal form representation of this game and answer the given subquestions.

Sub questions

Question Number : 452 Question Id : 640653904524 Question Type : MCQ Calculator : Yes Correct Marks : 1

Question Label : Multiple Choice Question

If player 1 chooses 1.5 hour as his action, the best response of player 2 will be

Options :

6406533045567. ✖ 1.5

6406533045568. ✖ 1

6406533045569. ✔ 1.25

6406533045570. ✖ 0

Question Number : 453 Question Id : 640653904525 Question Type : MCQ Calculator : Yes Correct Marks : 1

Question Label : Multiple Choice Question

How much negative campaign time each player will choose under a symmetric and unique Nash equilibrium?

Options :

6406533045571. ✖ 1.5

6406533045572. ✔ 1

6406533045573. ✖ 1.25

6406533045574. ✖ 0

Question Number : 454 Question Id : 640653904526 Question Type : MCQ Calculator : Yes Correct Marks : 1

Question Label : Multiple Choice Question

Equilibrium payoff of each player will be

Options :

6406533045575. ✖ -1.5

6406533045576. ✖ 1

6406533045577. ✔ -1

6406533045578. ✖ 0

Question Number : 455 Question Id : 640653904527 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

Now assume that this game is repeated infinitely. If the parties cooperate and sign a binding agreement on how much to campaign, they choose

$$a_1 = a_2 = 0.$$

What will be the payoff of each player in each stage game if they stick to this agreement

Options :

6406533045579. ✖ -1.5

6406533045580. ✖ 1

6406533045581. ✖ -1

6406533045582. ✔ 0

Question Number : 456 Question Id : 640653904528 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

Now assume that this game is repeated infinitely. If the parties cooperate and sign a binding agreement on how much to campaign, they choose

$$a_1 = a_2 = 0.$$

What should be the level of campaign a player should choose if he wants to deviate from this agreement to get the payoff of $\frac{1}{4}$

Options :

6406533045583. ✖ 1

6406533045584. ✔ 1/2

6406533045585. ✖ 1/3

6406533045586. ✖ 1/4

Question Number : 457 Question Id : 640653904529 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

Now assume that this game is repeated infinitely. If the parties cooperate and sign a binding agreement on how much to campaign, they choose

$$a_1 = a_2 = 0.$$

For which discount factors $\delta \in (0, 1)$ can this agreement cannot be supported as a subgame perfect equilibrium of the infinitely repeated game?

Options :

6406533045587. ✖ $\delta \geq \frac{1}{5}$

6406533045588. ✔ $\delta \leq \frac{1}{5}$

6406533045589. ✖ $\delta \geq \frac{1}{3}$

6406533045590. ✖ $\delta \leq \frac{1}{3}$